Chapter 7 — Waves and Vibrations

Board-style MCQs · PECTAA Physics 11 · Answer key at end

Practice MCQs

1. A wave transfers:

(a) neither **(b)** both matter and energy **(c)** matter without energy **(d)** energy without matter

2. Sound waves in air are:

(a) transverse **(b)** stationary only **(c)** electromagnetic **(d)** longitudinal

3. Light waves are:

(a) longitudinal **(b)** sound **(c)** transverse **(d)** mechanical

4. The wave equation is:

(a) $v = \lambda/f$ **(b)** $v = f + \lambda$ **(c)** $v = f/\lambda$ **(d)** $v = f\lambda$

5. Intensity is proportional to:

(a) amplitude **(b)** wavelength only **(c)** amplitude squared **(d)** frequency only

6. If amplitude is tripled, energy increases by:

(a) 6 times **(b)** 9 times **(c)** 12 times **(d)** 3 times

7. The principle of superposition gives resultant displacement as:

(a) difference of frequencies **(b)** sum of wavelengths **(c)** vector sum of displacements **(d)** product of displacements

8. Coherent sources have:

(a) zero amplitude **(b)** constant phase difference **(c)** random phase **(d)** different frequencies always

9. Constructive interference occurs for path difference:

(a) $\lambda/4$ **(b)** $(2n+1)\lambda/2$ **(c)** $n\lambda$ **(d)** $\lambda/8$

10. Destructive interference occurs for path difference:

(a) $(2n+1)\lambda/2$ **(b)** $2n\lambda$ **(c)** $n\lambda/4$ **(d)** $n\lambda$

11. Noise-cancelling headphones use:

(a) refraction **(b)** diffraction **(c)** destructive interference **(d)** constructive interference

12. Stationary waves are formed by:

(a) one wave only **(b)** two same-direction waves **(c)** random waves **(d)** two opposite identical waves

13. In a stationary wave, nodes are points of:

(a) maximum displacement **(b)** maximum speed **(c)** zero wavelength **(d)** zero displacement

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14. Antinodes are points of:

(a) zero frequency **(b)** zero displacement **(c)** zero amplitude **(d)** maximum displacement

15. Distance between adjacent nodes is:

(a) λ/4 **(b)** λ/2 **(c)** 2λ **(d)** λ

16. Distance between a node and next antinode is:

(a) λ/4 **(b)** 2λ **(c)** λ/2 **(d)** λ

17. No energy flows past a:

(a) trough **(b)** antinode **(c)** node **(d)** crest

18. For a string fixed at both ends, fundamental frequency is:

(a) $v/4\ell$ **(b)** $v/2\ell$ **(c)** $2v/\ell$ **(d)** v/ℓ

19. For a string, wave speed is:

(a) μ/F **(b)** √(μ/F) **(c)** √(F/μ) **(d)** F/μ

20. Increasing string tension:

(a) does not change frequency **(b)** decreases frequency **(c)** stops wave **(d)** increases frequency

21. An open pipe has fundamental:

(a) v/ℓ **(b)** $v/2\ell$ **(c)** $v/4\ell$ **(d)** $2v/\ell$

22. A closed pipe has fundamental:

(a) $v/2\ell$ **(b)** $2v/\ell$ **(c)** v/ℓ **(d)** $v/4\ell$

23. An open pipe produces:

(a) even harmonics only **(b)** odd harmonics only **(c)** all harmonics **(d)** no harmonics

24. A closed pipe produces:

(a) all harmonics **(b)** odd harmonics only **(c)** no harmonics **(d)** even harmonics only

25. A closed pipe of same length sounds:

(a) higher than open pipe **(b)** silent **(c)** same as open pipe **(d)** lower than open pipe

26. Diffraction is:

(a) bending in medium **(b)** superposition **(c)** bouncing back **(d)** spreading around obstacles

27. Diffraction is most noticeable when gap is:

(a) much larger than wavelength **(b)** infinite **(c)** zero **(d)** comparable to wavelength

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28. Beats are produced by:

(a) same frequency only **(b)** stationary waves only **(c)** slightly different frequencies **(d)** zero amplitude

29. Beat frequency equals:

(a) $f_1 f_2$ **(b)** $|f_1 - f_2|$ **(c)** $f_1 + f_2$ **(d)** $(f_1 + f_2)/2$

30. Beats become hard to recognise above about:

(a) 1 Hz **(b)** 100 Hz **(c)** 1000 Hz **(d)** 10 Hz

31. Loading a tuning fork with wax:

(a) raises its frequency **(b)** lowers its frequency **(c)** stops it always **(d)** does not change frequency

32. The Doppler effect is change in:

(a) wave speed in medium **(b)** observed frequency due to relative motion **(c)** amplitude only **(d)** wavelength only in vacuum

33. A source moving toward a stationary listener gives:

(a) higher frequency **(b)** same frequency **(c)** zero frequency **(d)** lower frequency

34. For source moving toward listener, f′ equals:

(a) f(v − u_s)/v **(b)** vf/(v − u_s) **(c)** vf/(v + u_s) **(d)** f(v + u_s)/v

35. For source moving away, f′ equals:

(a) f(v + u_s)/v **(b)** vf/(v + u_s) **(c)** vf/(v − u_s) **(d)** f(v − u_s)/v

36. A receding star shows:

(a) blue shift **(b)** red shift **(c)** green shift **(d)** no shift

37. Doppler effect is used in:

(a) radar speed traps **(b)** only balances **(c)** only microscopes **(d)** only thermometers

38. Stationary waves oscillate:

(a) with changing velocity **(b)** with constant velocity **(c)** through vacuum only **(d)** in fixed positions

39. Harmonics are:

(a) integer submultiples **(b)** integer multiples of fundamental **(c)** non-integer multiples **(d)** random frequencies

40. Constructive interference gives:

(a) no amplitude **(b)** zero frequency **(c)** decreased amplitude **(d)** increased amplitude

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41. Headphones produce sound using:

(a) transverse waves in air **(b)** stationary light **(c)** electromagnetic waves only **(d)** longitudinal pressure waves

42. Typical microwave frequency range is:

(a) 10^2–10^4 Hz **(b)** 10^6–10^8 Hz **(c)** 10^{14}–10^{16} Hz **(d)** 10^9–10^{11} Hz

43. Bending of waves around an obstacle is:

(a) reflection **(b)** refraction **(c)** interference **(d)** diffraction

44. Doppler effect in astronomy determines:

(a) temperature of sun only **(b)** velocity of galaxies **(c)** diameters of stars only **(d)** mass of planets only

45. The most important measurement for intensity is:

(a) amplitude **(b)** speed only **(c)** frequency only **(d)** wavelength only

46. Two identical waves travelling in same direction:

(a) form nodes only **(b)** always form stationary waves **(c)** form beats only **(d)** do not form stationary waves

47. Compressions and rarefactions correspond to:

(a) frequency and wavelength **(b)** crests and troughs **(c)** speed and velocity **(d)** nodes and antinodes

48. If source moves perpendicular to observer line, frequency:

(a) decreases **(b)** increases **(c)** does not change **(d)** becomes zero

49. Ripple tank is used to study:

(a) only heat **(b)** only sound **(c)** only light speed **(d)** diffraction and interference

50. The intensity of a progressive wave is:

(a) force per unit area **(b)** power per unit area **(c)** energy per unit volume only **(d)** momentum per unit area

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Answer Key with Brief Reasons

1. Answer: (d) energy without matter — Particles oscillate about mean positions.

2. Answer: (d) longitudinal — Air particles vibrate parallel to travel.

3. Answer: (c) transverse — Light has perpendicular vibrations.

4. Answer: (d) v = fλ — Speed = frequency × wavelength.

5. Answer: (c) amplitude squared — $I \propto A^2$.

6. Answer: (b) 9 times — Energy $\propto A^2$.

7. Answer: (c) vector sum of displacements — Displacements add.

8. Answer: (b) constant phase difference — Stable interference needs coherence.

9. Answer: (c) nλ — Waves arrive in phase.

10. Answer: (a) (2n+1)λ/2 — Waves arrive out of phase.

11. Answer: (c) destructive interference — They produce opposite-phase sound.

12. Answer: (d) two opposite identical waves — Superposition of opposite waves.

13. Answer: (d) zero displacement — Nodes remain at rest.

14. Answer: (d) maximum displacement — Antinodes oscillate most.

15. Answer: (b) λ/2 — Nodes are half wavelength apart.

16. Answer: (a) λ/4 — Quarter wavelength.

17. Answer: (c) node — Node remains at rest.

18. Answer: (b) v/2ℓ — $\lambda_1 = 2\ell$.

19. Answer: (c) √(F/μ) — F tension, μ mass per unit length.

20. Answer: (d) increases frequency — $v = \sqrt{(F/\mu)}$.

21. Answer: (b) v/2ℓ — Both ends antinodes.

22. Answer: (d) v/4ℓ — One node, one antinode.

23. Answer: (c) all harmonics — $f_n = nv/2\ell$.

24. Answer: (b) odd harmonics only — $f_n = nv/4\ell$, n odd.

25. Answer: (d) lower than open pipe — Fundamental is half.

26. Answer: (d) spreading around obstacles — Waves spread through gaps.

27. Answer: (d) comparable to wavelength — Strong spreading occurs.

28. Answer: (c) slightly different frequencies — Periodic loudness variation.

29. Answer: (b) $|f_1 - f_2|$ — Difference of frequencies.

30. Answer: (d) 10 Hz — Ear cannot follow rapid variations.

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31. Answer: (b) lowers its frequency — Added mass lowers frequency.

32. Answer: (b) observed frequency due to relative motion — Relative motion changes frequency.

33. Answer: (a) higher frequency — Wavefronts are crowded.

34. Answer: (b) $vf/(v - u_s)$ — Denominator decreases.

35. Answer: (b) $vf/(v + u_s)$ — Denominator increases.

36. Answer: (b) red shift — Frequency decreases.

37. Answer: (a) radar speed traps — It measures relative speed.

38. Answer: (d) in fixed positions — Pattern does not travel.

39. Answer: (b) integer multiples of fundamental — $f_n = nf_1$.

40. Answer: (d) increased amplitude — Waves reinforce.

41. Answer: (d) longitudinal pressure waves — Sound in air is longitudinal.

42. Answer: (d) 10^9–10^{11} Hz — Microwaves are GHz range.

43. Answer: (d) diffraction — Diffraction spreads waves.

44. Answer: (b) velocity of galaxies — Red/blue shift gives speed.

45. Answer: (a) amplitude — $I \propto A^2$.

46. Answer: (d) do not form stationary waves — Stationary waves need opposite directions.

47. Answer: (b) crests and troughs — Longitudinal extremes match transverse extremes.

48. Answer: (c) does not change — No radial relative motion.

49. Answer: (d) diffraction and interference — Water waves show wave behaviour.

50. Answer: (b) power per unit area — $I = P/A$.

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